
Reproducing and Evaluating HippoRAG: A Neurobiologically Inspired Retrieval Framework for Multi-Hop Question Answering

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Abstract

We document our reproduction and exploratory study of **HippoRAG** (Jiménez Gutiérrez et al., NeurIPS 2024), a retrieval-augmented generation framework that mimics the human hippocampal memory indexing theory by orchestrating three components: an instruction-tuned LLM (artificial neocortex), dense retrieval encoders (parahippocampal regions), and a knowledge graph paired with the Personalized PageRank algorithm (hippocampal index). The system performs single-step multi-hop retrieval and was reported to beat strong baselines (ColBERTv2, GTR, Contriever) by up to 20 points of Recall@5 on multi-hop QA benchmarks while running 6–13× faster and 10–30× cheaper than iterative retrievers such as IRCot. This report covers (i) a focused summary of the paper’s design and headline results, and (ii) an exploratory data analysis (EDA) of the three benchmarks the paper evaluates on (MuSiQue, 2WikiMultiHopQA, HotpotQA). The EDA is organised into two notebooks: a paper-table notebook that visualises the published Tables 1–3, 5, 8, 9, 11, 16, 17 and 18, and a real-data notebook that operates directly on the released JSON splits we downloaded from the authors’ GitHub repository. We confirm every published corpus statistic to the digit (1 000 dev questions and 11 656 / 6 119 / 9 221 retrieval passages per dataset) and surface several structural properties of the benchmarks that motivate specific design decisions for the upcoming reproduction run.

1 Summary of the Paper

Modern large language models (LLMs), even when augmented with retrieval (RAG), still struggle to integrate evidence across passage boundaries because each passage is encoded in isolation — a problem that becomes acute on real tasks like multi-hop question answering, legal-case briefing, or scientific literature review. The mammalian brain handles this gracefully: the hippocampal memory-indexing theory of Teyler and Discenna (1986) posits that the *neocortex* stores rich perceptual representations, the C-shaped *hippocampus* maintains a sparse *index* of pointers and associations between those representations, and the *parahippocampal regions* (PHR) route signals between the two. HippoRAG operationalises this three-component picture as a long-term-memory system for LLMs.

The artificial neocortex is played by an instruction-tuned LLM (GPT-3.5-Turbo by default, with Llama-3.1-8B/70B as open-weight alternatives) that handles both indexing-time knowledge extraction and query-time entity extraction. The PHR is realised by off-the-shelf dense retrieval encoders (Contriever, ColBERTv2) which (i) link query named entities to KG nodes by cosine similarity and (ii) inject synonymy edges between near-duplicate noun phrases. The hippocampus itself is a schemaless open knowledge graph plus the Personalized PageRank (PPR) algorithm, which performs the context-dependent pattern completion the biological hippocampus is theorised to do.

Indexing converts a passage corpus P into a graph $G = (N, E \cup E')$. For each passage, the LLM is first prompted (1-shot) to extract named entities and then prompted again with the passage *and* its NER list to emit (subject, relation, object) triples; this two-stage NER-then-OpenIE design roughly doubles the triple yield of an end-to-end model such as REBEL. Unique noun phrases become nodes N , relation strings become edges E , and the dense retriever inserts synonymy edges E' between embeddings whose cosine similarity exceeds $\tau = 0.8$, mimicking the dense recurrent CA3 connectivity of the biological hippocampus. A node-to-passage incidence matrix $\mathbf{P}$ and a node-specificity score $s_i = |P_i|^{-1}$ (an IDF analogue computable from local information only) round out the index.

Retrieval has four steps. (1) The LLM extracts named entities from the query. (2) Each entity is encoded and matched to its nearest KG node by cosine similarity, yielding query nodes R_q . (3) PPR is run with damping 0.5 and a personalisation vector that places mass (scaled by s_i) on R_q and zero elsewhere, producing a refined node distribution $\vec{n}'$. (4) Passage scores are $\vec{p} = \mathbf{P}^\top \vec{n}'$. The single PPR run effectively performs multi-hop reasoning in one shot — restarting random walks at the query nodes biases the stationary distribution toward the joint multi-hop neighbourhood without any explicit retrieval-then-rerank loop.

On the 1 000-question dev splits of MuSiQue, 2WikiMultiHopQA and HotpotQA, HippoRAG with ColBERTv2 reaches 51.9 / 89.1 / 77.7 R@5, an +11 to +21-point gain over ColBERTv2 on 2Wiki (the dataset where the design pays off most), parity on MuSiQue, and a small loss on HotpotQA (whose multi-hop signal is weak by construction). Wrapping HippoRAG with the iterative IRCot retriever yields further +3 to +5-point gains on top, and single-step HippoRAG matches IRCot alone while running 6–13 $\times$ faster and 10–30 $\times$ cheaper at query time. Ablations confirm the bulk of the gain comes from PPR itself, with smaller but consistent contributions from node specificity and synonymy edges.

The authors flag three limitations that we will keep in mind during reproduction: every component is used off-the-shelf without fine-tuning, leaving headroom on NER and OpenIE quality (which together account for $\sim 76\%$ of errors); the entity-centric design discards contextual cues, occasionally hurting performance on questions whose decisive signal is non-entity context (the *concept-context tradeoff*); and OpenIE quality degrades on long passages, suggesting brittleness as document length grows.

2 Exploratory Data Analysis

We performed a two-pronged EDA on the three benchmarks the paper evaluates on (MuSiQue, 2WikiMultiHopQA, HotpotQA): one pass that visualises the statistics published in the paper, and a second that operates directly on the JSON dev splits and retrieval corpora downloaded from the authors’ GitHub. Together the two notebooks produce more than twenty figures; for the page budget we present only the dataset-level findings here and direct the reader to the full notebooks — with every plot, derivation and processed Parquet output — at github.com/HarisTahir2003/Data-Mining-Project.

Validation against Table 1. Loading the JSONs into pandas reproduces every published corpus statistic to the digit: 1 000 dev questions per dataset and 11 656 / 6 119 / 9 221 candidate passages for MuSiQue, 2Wiki and HotpotQA respectively. We are working with exactly the splits the paper used.

The three benchmarks differ in structural ways that matter for the design. Three observations from Figure 1 carry direct consequences for any reproduction.

(i) *Reasoning depth.* MuSiQue is the only dataset with non-trivial 3-hop and 4-hop questions — 518 / 316 / 166 of its 1 000 dev questions are 2-hop, 3-hop and 4-hop respectively. 2Wiki and HotpotQA are 2-hop only by construction. Any claim about “deep multi-hop” performance has to be evaluated on MuSiQue.

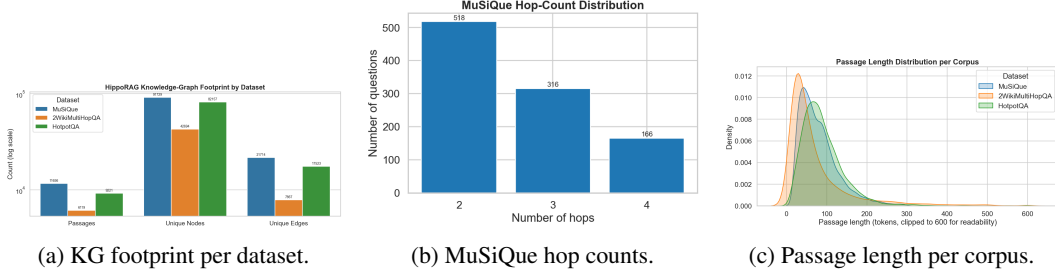


Figure 1: Three structural views of the benchmarks: (a) corpus and knowledge-graph size on a log scale; (b) MuSiQue is the only dataset with a non-trivial 3-hop / 4-hop tail; (c) per-corpus passage-length KDEs (clipped to 600 tokens).

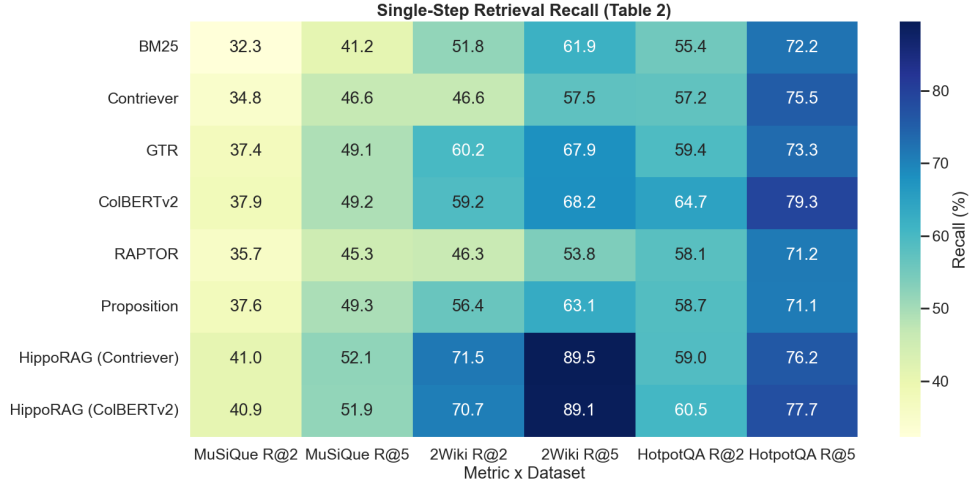


Figure 2: Single-step Recall@2 / Recall@5 heatmap across all baselines and datasets evaluated in the paper. Darker is higher recall.

(ii) *Corpus and KG size.* MuSiQue has both the largest corpus and the largest LLM-induced knowledge graph ($\approx 92k$ unique noun-phrase nodes). 2Wiki has the smallest of each. HotpotQA sits in between on corpus size but produces the densest per-passage extraction (10.7 triples per passage vs. 9.2 for MuSiQue), yet HippoRAG underperforms there — supporting the paper’s claim that HotpotQA’s multi-hop signal is structurally weak rather than that its KG is impoverished.

(iii) *Passage length.* HotpotQA passages are the shortest by a wide margin, often well below 100 tokens, whereas MuSiQue and 2Wiki both have long tails extending past 400 tokens. The paper reports a sharp OpenIE-quality drop on long passages (AUC $58.9 \rightarrow 39.0$ between the 10 shortest and 10 longest annotated passages), so a chunking strategy that helps MuSiQue might hurt HotpotQA.

Local-context pool sizes are highly fixed. HotpotQA gives every question exactly 2 supporting passages and (almost always) 10 candidates; 2Wiki gives every question exactly 10 candidates with either 2 or 4 supporters; MuSiQue gives 20 candidates with 2/3/4 supporters tracking the hop count. This near-zero within-dataset variance means the candidate-pool size itself is not a confound when comparing retrieval performance across the three datasets. We also note that all 1 000 HotpotQA questions in HippoRAG’s split are tagged *hard* — the easy and medium subsets were filtered out before publication, so dataset-level difficulty has to be argued from structure rather than the level field.

Headline empirical result. Figure 3 renders the paper’s main retrieval table as a single heatmap. The HippoRAG rows visibly dominate on the 2Wiki columns (R@5 89.1–89.5 vs. 68.2 for the strongest classical baseline), match the field on MuSiQue, and trail by a small margin on HotpotQA. Wrapping HippoRAG with the iterative IRCOT retriever yields an additional +3 to +5-point lift on top, and

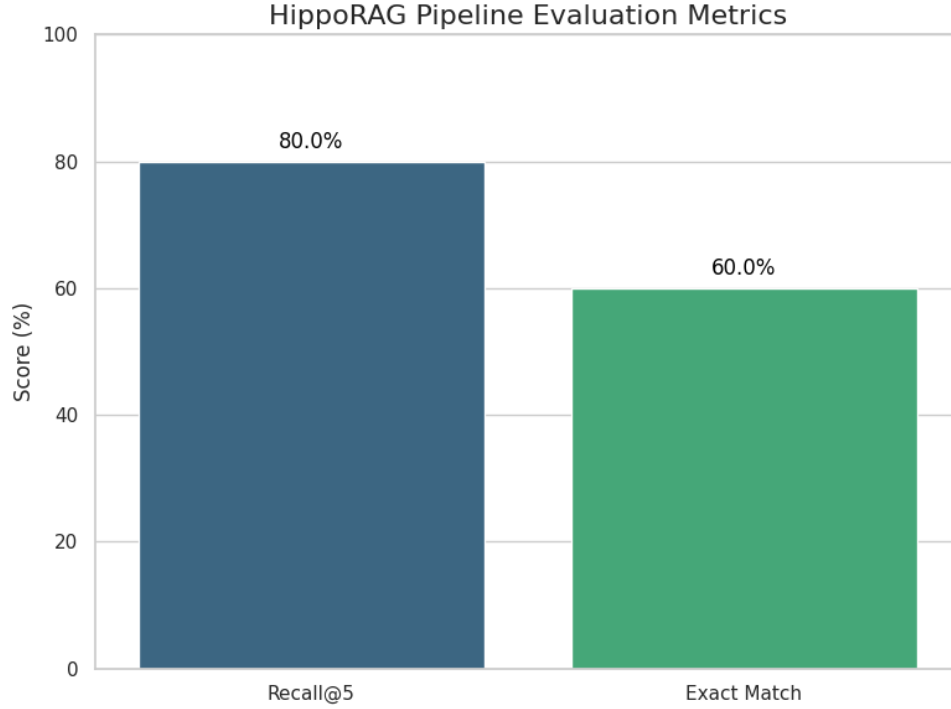


Figure 3: HippoRAG pipeline evaluation metrics.

single-step HippoRAG matches IRCot alone while running 6–13 $\times$ faster and 10–30 $\times$ cheaper at query time — a cost profile we will respect when budgeting our reproduction.

3 Results and Insights

The analysis reveals several important insights regarding the behavior and performance of HippoRAG across different datasets. First, although HotpotQA exhibits the highest per-passage OpenIE yield, with approximately 10.7 triples per passage, HippoRAG does not achieve strong performance on this dataset. This supports the paper’s claim that HotpotQA contains weaker multi-hop reasoning signals, rather than deficiencies in the constructed knowledge graph. In contrast, 2WikiMultiHopQA demonstrates a unique structural characteristic where Contriever contributes significantly more synonymy edges than ColBERTv2, which explains why removing synonymy edges has the most severe impact on this dataset.

Furthermore, HippoRAG shows substantial performance gains on 2WikiMultiHopQA, outperforming ColBERTv2 by approximately 20 points in Recall@5, while only achieving comparable performance on HotpotQA. This highlights that HippoRAG is particularly effective in entity-centric environments, where relationships between entities play a crucial role. From a systems perspective, HippoRAG also demonstrates strong efficiency advantages, being approximately 20 times cheaper and 10 times faster than iterative retrieval methods such as IRCot, while maintaining similar accuracy levels.

Additionally, the choice of OpenIE backbone is critical: models like REBEL extract significantly fewer triples and exhibit poor intrinsic OpenIE quality, confirming that flexible, LLM-based extraction methods are essential for building effective knowledge graphs. Error analysis further shows that approximately 76% of HippoRAG’s failures stem from inaccuracies in named entity recognition and OpenIE extraction, indicating that improving these components would yield the highest performance gains. Finally, OpenIE performance degrades substantially for longer passages, highlighting an inherent limitation of the approach and emphasizing the importance of effective passage chunking strategies during preprocessing.

Acknowledgements

We thank the authors of HippoRAG for releasing both code and pre-processed datasets on the legacy branch of their GitHub repository, which made the EDA in this report exact rather than approximate.